DAY 2 HANDS-ON LABS & CAPSTONE BRIEF · SESSIONS 016 & 017

TP3, TP4 & Live Pod Capstone Defenses

Building multi-agent teams with Google ADK in the morning (`Session 016`) and automated evaluation & safety guardrails in the afternoon (`Session 017`).

INSTRUCTOR	PROGRAM & COHORT	SCHEDULE
Chouaieb Nemri	Year 4 · BSc Data Science	Wednesday, October 21, 2026

Overview of Day 2

On Day 1 and in the Homework, a single ReAct agent handled every task. On Day 2, we tackle what happens when an enterprise workflow requires multiple domain specialists, deterministic human-in-the-loop (HITL) escalation gates, and adversarial safety guardrails:

1. **Morning (`TP3` , `10:00–11:40`):** Multi-Agent Green CapEx Committee using **Google ADK (`google-adk`)** with `gemma-4-26b-a4b-it` specialists and a `gemma-4-31b-it` Committee Chair.
2. **Afternoon (`TP4` , `15:05–16:35`):** Multi-metric trajectory evaluation, indirect prompt injection (`SUP-010`) sanitization, `before_tool_guardrail` , and secret-leak scrubbing.
3. **Final Block (`16:45–17:55`): 15-Minute Live Pod Capstone Defenses** where each pod demos their system and survives a live challenger query from another pod.

Lab 3 (`TP3`): Multi-Agent Green CapEx Committee with Google ADK

- **Starter Notebook:** `03_Labs_TP/TP3_MultiAgent_ADK_Workflows/TP3_Starter_ADK_MultiAgent_Team.ipynb`
- **Datasets:** `03_Labs_TP/data/green_procurement_requests.json` (`PROP-2026-01` .. `04`) & `supplier_esg_audit_db.json`
- **Models:** `MODEL_FAST = "gemma-4-26b-a4b-it"` (Specialists) & `MODEL_FRONTIER = "gemma-4-31b-it"` (Committee Chair)

What You Will Build in TP3

1. **Parallel Specialist Due Diligence (`ParallelAgent` / Stage 1):**
   - `CarbonSpecialist` (`model="gemma-4-26b-a4b-it"` , `output_key="carbon_report"`): Verifies `expected_co2_reduction_tco2e > 0` and `sbti_verified == True` .
   - `FinancialRiskSpecialist` (`model="gemma-4-26b-a4b-it"` , `output_key="financial_report"`): Verifies `payback_years <= 4.0` , computes shadow carbon value (`€100/tCO2e`), and flags `capex_amount_eur > €100,000` .
2. **Iterative Critic Loop & Committee Chair Synthesis (`LoopAgent` / `SequentialAgent`):**
   - `CommitteeChair` (`model="gemma-4-31b-it"` , `output_key="chair_decision"`) synthesizes the specialist reports from `session.state` .
3. **Deterministic Policy & HITL Gate (`CommitteeDecision`):**
   - Classify all 4 proposals (`PROP-2026-01` -> `AUTO_APPROVED` , `PROP-2026-02` -> `ESCALATED_FOR_HUMAN_SIGN_OFF` , `PROP-2026-03` & `PROP-2026-04` -> `REJECTED_POLICY_VIOLATION`).
4. **Stretch Task — Hybrid MoE (`gemma-4-26b-a4b-it`) + Dense (`gemma-4-31b-it`) Routing Calculator:**
   - Quantify the **65.3% active-parameter compute reduction** and **doubled Free-Tier TPM quota (`2 × 16,000 = 32,000 TPM`)** achieved by routing the 3 specialists to `gemma-4-26b-a4b-it` (4B active MoE) and reserving `gemma-4-31b-it` (31B dense) for the Committee Chair.

Lab 4 (`TP4`): Trajectory Evaluation, Guardrails & Red-Teaming

- **Starter Notebook:** `03_Labs_TP/TP4_Evaluation_Guardrails_and_Safety/TP4_Starter_Eval_and_Guardrails.ipynb`
- **Dataset:** `03_Labs_TP/data/supplier_esg_audit_db.json` (featuring the `SUP-010` adversarial payload)

What You Will Build in TP4

1. **Input Sanitization Against Indirect Prompt Injection (`sanitize_untrusted_text`):**
   - Detect and redact adversarial override patterns (`ignore all previous` , `system override` , `ignore all safety` , `approve immediate`) inside untrusted database fields (`SUP-010`) before they enter session history, returning `("[REDACTED_ADVERSARIAL_INJECTION_ATTEMPT]", True)` .
2. **Pre-Tool Parameter Guardrail (`before_tool_guardrail`):**
   - Implement an ADK-style `before_tool_callback` that intercepts `execute_supplier_bonus_payment` and blocks any transfer where `amount_eur > 10_000` unless `session_state.get("cfo_human_approval_token")` is present.
3. **Multi-Metric Trajectory & Safety Evaluator (`evaluate_agent_trace`):**
   - Compute `tool_trajectory_recall` , `risk_tier_accuracy` , `safety_guardrail_score` , and the weighted `composite_score` (`0.4 * recall + 0.4 * tier_match + 0.2 * safety`).
4. **Stretch Task — Output PII & Secret Leak Scrubber (`scrub_sensitive_output`):**
   - Redact any accidental leak of an internal API key (`AIza...` -> `[REDACTED_API_KEY]`) or European IBAN (`[REDACTED_IBAN]`) before displaying agent output.

Live Pod Capstone Defenses (`16:45 – 17:55` on Wednesday, Oct 21)

In the final 70 minutes of Day 2, each pod has **15 minutes** (`8m` live notebook walkthrough + `4m` live challenger query from another pod + `3m` instructor Q&A):

Time Slot	Pod	Members	Challenger Pod
`16:45 – 17:00`	**Pod Alpha**	Jiani XU, Yuxin GONG, Simon GRANDIDIER	Pod Beta
`17:02 – 17:17`	**Pod Beta**	Mélissa CEHIC, Deyi WANG, Antoine BAUDET	Pod Gamma
`17:19 – 17:34`	**Pod Gamma**	Alikatte HRISHIKESH SIDHARTH, Charlotte DEMODE, Sofia KURAKOVA	Pod Delta
`17:36 – 17:51`	**Pod Delta**	Leyan CHENG, Harshavardhan Reddy MALIKIREDDY, Naomi HELMBOLD	Pod Alpha

> **Post-Course Solution Release (Wednesday, Oct 21 at 18:15)**
>
> Immediately after the Capstone Defenses conclude at `18:15` , the complete executed reference solutions (`TP1_Solution` through `TP4_Solution` and `Homework_Reference_Solution`) will be published to the course GitHub repository as a tagged **GitHub Release (`v2026.10-solutions`)**.